Assault XLII-S7.12: fourierCos as a True Definition and the Scale Lemma fourierCos_scale as a True Proof

(Integral/analytic-layer component · 2026-10-29 · archived to d1 by convention)

1 Summary of Conclusions

This round completes component 5 of the integral/analytic layer of the QRH Lean formalization project (C:\proj2\QRH): the frequency-domain operator fourierCos is demoted from axiom to a Bochner-integral true definition, and its scale law fourierCos_scale (r̂ _a(z) = r̂ (az)) is demoted from axiom to a fully proved lemma. The full lake build QRH passes: 2704 jobs, 0 errors, 0 warnings.

One earlier conclusion is corrected. The S7.3 documentation and code comments claimed that "a true proof of fourierCos_scale requires an integral layer (MeasureTheory.IntervalIntegral or a Riemann integral module), to be restored after the mathlib aggregate is generated." This round refutes that judgment empirically: mathlib v4.34 has restructured Bochner integration into the Integral/Bochner/ subdirectory, and the full-space Haar change-of-variables theorem Measure.integral_comp_inv_mul_left is already usable under the existing "specific-module import" route, with no regularity assumption on the integrand. The scale law therefore closes directly, with no module aggregation needed.

The count of remaining named axioms decreases by one (fourierCos_scale leaves); the two dependent use sites n1_diag_invariant and n1_offdiag_scale_eq are unchanged in statement and compile without modification.

2 Background and Positioning

2.1 Role of fourierCos in the S7.1 Formalization

fourierCos is the frequency-domain abstract operator: for a pairing kernel K and a frequency point z it gives $\int K(x)\cdot\cos(2\pi zx)\,dx$, corresponding to the kernel Fourier transform r̂ (z) in the spectral-duality framework. In the S7.1 theorem skeleton the upper-bound chain for the pairing constant A_L is carried through the L^1 bound and the frequency-decay component of the kernel Fourier transform; the N1 super-support structural components of S7.3, n1_diag_invariant (z = 0 specialization) and n1_offdiag_scale_eq (z = 2δ specialization), depend on rw [fourierCos_scale ...] as their only dependency.

2.2 Axiom Backlog Status

Core §4b originally carried three named axioms: fourierCos (the carrier), fourierCos_Khat (closed form of K̂), and khat_lipschitz (Lipschitz component of the band-limited entire function); Stage3 N1 region further carried fourierCos_scale (frequency scaling). This round removes fourierCos itself and fourierCos_scale from the backlog; fourierCos_Khat and khat_lipschitz remain axioms (the former needs the analytic closed-form layer, the latter the derivative component; separate assault lines).

3 Change 1: fourierCos as a True Definition

In Core §4b the axiom fourierCos is replaced by a noncomputable true definition:
fourierCos K z := ∫ x : ℝ, K(x)·cos(2π·z·x) dx

Semantics and convention: the integral is a Bochner integral with respect to volume on ℝ; when the integrand is not integrable, the mathlib convention returns 0 (the same mechanism as the true definition of normL1). The calling signature fourierCos K z is unchanged, so all use sites — the primeSum weights, prime_weight_bounds_Khat, and others — need no modification. The comments are updated to the true-definition statement.

4 Change 2: The Scale Lemma fourierCos_scale as a True Proof

4.1 Statement

For arbitrary r : ℝ → ℝ (no regularity assumption), 0 < a, z : ℝ:
fourierCos (scaleKernel a r) z = fourierCos r (a·z)

where scaleKernel a r := λx, (1/a)·r(x/a).

4.2 Proof Chain

A four-step calc chain (same technology stack as the scaleKernel_L1_invariant proof of 2026-10-27):

(1) Pull out the constant factor: the integrand (1/a)·r(x/a)·cos(2πzx) is rearranged by integral_congr_ae + ring into (1/a)·(r(x/a)·cos(2πzx)), then integral_const_mul extracts 1/a.

(2) Rearrange the integrand: x/a is rewritten as $a^{-1}\cdot x$ (div_eq_inv_mul), and the cos argument is rewritten accordingly as $2\pi\cdot z\cdot(a\cdot(a^{-1}\cdot x))$ — the latter requires field_simp [ne_of_gt ha] to cancel a^{-1}, the only field-operation point of this step.

(3) Haar change of variables: call Measure.integral_comp_inv_mul_left directly, obtaining $\int r(a^{-1}\cdot x)\cdot\cos(2\pi\cdot z\cdot(a\cdot(a^{-1}\cdot x)))\,dx = |a|\cdot\int r(y)\cdot\cos(2\pi\cdot z\cdot(a\cdot y))\,dy$. This theorem is the full-space version and holds for arbitrary functions with no measurability assumption.

(4) Rewrite the cos argument and cancel the factor: $z\cdot(a\cdot y)$ is rewritten as $(a\cdot z)\cdot y$ (integral_congr_ae + ring), and the final term $(1/a)\cdot(|a|\cdot X)$ is reduced to X by abs_of_pos ha and field_simp [ne_of_gt ha], which is exactly fourierCos r $(a\cdot z)$.

4.3 Available Components and Use Sites

Available: integral_const_mul, integral_congr_ae (filter_upwards pointwise architecture), and Measure.integral_comp_inv_mul_left (full space, no regularity assumption) — all located in Integral/Bochner/Basic.olean, inside the project's existing import chain. The use sites n1_diag_invariant (rw [fourierCos_scale a ha r 0]) and n1_offdiag_scale_eq (rw [fourierCos_scale a ha r $(2\cdot\delta)$]) are unchanged in signature and compile as before.

5 Correction of an Earlier Conclusion: the IntervalIntegral Dependency Judgment

5.1 The Earlier Judgment

The S7.3 archive and code comments recorded: "a true proof of fourierCos_scale requires an integral layer (MeasureTheory.IntervalIntegral or a Riemann integral module) — to be restored after the mathlib aggregate is generated," based on probes confirming that IntervalIntegral.olean and the top-level Integral.olean are missing.

5.2 Empirical Refutation

(a) The missing-judgment part is correct: IntervalIntegral.olean, the top-level Integral.olean, and the legacy name Bochner.olean all do not exist.

(b) The conclusion was wrong: v4.34 restructures Bochner integration into the Integral/Bochner/ subdirectory; the $\int$ notation, integral_const_mul, integral_congr_ae, and integral_map all live in Bochner/Basic.olean (the same source as the true definition of normL1, already inside the project import chain); moreover Measure.integral_comp_inv_mul_left is a full-space Haar change of variables with no regularity assumption on the integrand. The scale-law proof therefore needs neither IntervalIntegral nor any mathlib aggregation.

5.3 Dead Ends (Do Not Retry)

(a) import Mathlib.MeasureTheory.Integral.Integral: the top-level olean is missing; it always fails.

(b) A full-space pointwise integral_congr: does not exist in v4.34 (only integral_congr_ae and the L1-class versions); always use integral_congr_ae + filter_upwards / ae_of_all.

(c) Generalizing integral_map: that theorem requires AEMeasurable φ and AEStronglyMeasurable f (the integrand), unusable for arbitrary r; the measure-layer scaling formulas volume_map_mul_left / map_mul_left do not exist in v4.34. The only reliable exit for a full-space change of variables is Measure.integral_comp_inv_mul_left.

6 Build Verification

Full lake build QRH: 2704 jobs, exit = 0, 0 errors, 0 warnings (log C:\proj2\QRH\dbg_build_fourier.txt). The three probe versions (FourierProbe.lean: import-missing check, namespace-trap check, integral-component inventory check) completed their mission and were deleted.

7 Remaining Named Axiom Status

Name	Location	Status
fourierCos_Khat	Core §4b	Named axiom (closed form of K̂ , analytic layer)
khat_lipschitz	Core §4b	Named axiom (Lipschitz of band-limited entire function)
bgs5_khat_fourier_supp	Main	Named axiom (BGS Lemma 5 frequency support)
siegel_walfisz_chi5	Main	Named axiom (Siegel-Walfisz for χ_5)
landau_explicit_L	Main	Named axiom (Landau explicit formula, L version)
f0_sub_feps_L1_bound	Stage3	Named axiom (exact statement; true proof needs integral layer)
N_L_lower_bound	Stage3	Named axiom (structural; S6 zero count)
n2_linkage	Stage3	Named axiom (structural; S7.4 Theorem N2)
n3_remainder_uniform	Stage3	Named axiom (structural; S7.5 Theorem N3)

Note: the table is a snapshot as of 2026-10-29. fourierCos and fourierCos_scale have left the backlog; each remaining axiom belongs to its own assault line, as recorded in the respective chapters.